



Javier Díez Pérez

SENIOR DATA SCIENTIST & AI
INNOVATION SPECIALIST

- Tarragona, Spain
- +34 654 052 443
- jdiezperezj@gmail.com
- jdiez.github.io

CORE SKILLS

Programming

Python, R, SQL, Scala, Rust,
HTML/CSS

Data Science & ML

scikit-learn, XGBoost, PySpark,
Dask, Ray, H2O, SHAP, lime, Pandas,
Polars

AI & Agents

RAG/RIG, Multimodal RAG, Multi-
agent systems (Agno), LLM
chatbots, Knowledge Graphs

Knowledge Graphs

RDF/LPG, SPARQL, Neo4j,
Ontologies, Causal KGs, Biomedical
KGs, Data Modelling

Data Stores

PostgreSQL, MySQL, DuckDB,
MongoDB, Hive, Elasticsearch, RDF

Pipelines & CI/CD

Prefect, Airflow, Nextflow,
Snakemake, GitHub Actions, pytest,
Docker

Cloud & DevOps

AWS (Big Data & ML), GCP,
Terraform, Packer

Bioinformatics

Bioconductor, Biopython, NGS
pipelines, GATK, SAMTools, scanpy

Statistics & GIS

Experiment design, multivariate
analysis, PostGIS, QGIS

LANGUAGES

Spanish	Native
English	Proficient
French	Good
Portuguese	Basic

VOLUNTEERING

- CRAM Marine Animals Rehab
(2024–25)
- Madrid4Refugees coding teacher
- Save the Children tutor

PROFILE

Data scientist and AI engineer with 15+ years in bioinformatics, data engineering, and ML across pharma and biotech. Building agentic AI applications, multimodal RAG systems, and multi-agent pipelines (Agno) at AstraZeneca. Expertise in knowledge graphs (RDF, LPG, biomedical ontologies), data modelling, oncology, genomics, and health data platforms.

EXPERIENCE

- AstraZeneca · Barcelona

2023 — Present

Senior Innovation Specialist DS & AI (ODSAI, 2025–Present)

- Built agentic conversational app for structured/unstructured data (RAG/RIG) with multi-agent architecture using Agno framework
 - Developed multimodal RAG for enterprise knowledge retrieval
 - Alexion 2025 spring accelerator — innovation modelling via Strategyzer's framework

Data Product Lead — Oncology DS Platforms (2023–2025)

- Led data product operations for PDx mouse models and ENABL platform
 - Designed and maintained oncology data science platform infrastructure
- Data Engineer & Python Developer

2022 — 2023

Contractor · Madrid

- Data products for pharma (GSK) and food (HdR); Python packages for data validation & FAIRification
 - GCP ETL pipeline development with integrated validation framework
- Lead Bioinformatician & Data Scientist

2021 — 2022

PharmaMar · Madrid

- R&D bioinformatics lead: cancer genomics, drug development, biomarker discovery
 - Multimics (DNA-seq, RNA-seq, ChIP-seq, ATAC-seq) and clinical trial data integration
- Senior Bioinformatician & Data Scientist

2020 — 2021

Fujitsu · Madrid

- Digital transition at European CoE for Data Intelligence — EHR APIs, SNOMED-CT, entity recognition
- Senior Bioinformatics Scientist

2019 — 2020

HiFiBio · Paris

- Single cell RNA-seq analysis for TCR in TILs; sc-targeted RNAseq pipeline development
 - Algorithm development for cell barcode UMI assignment and sequence correction
 - Unsupervised ML data analysis for cell-type and TCR chain-type characterization
- Data Analyst & Python Developer

2017 — 2020

Freelancer · Madrid

- Predictive analytics tools for banking, insurance, and telco industries
 - Big data science and ML with Python and R ecosystem; package and tool development
- Senior Bioinformatician

2016

Wellcome Trust Sanger Institute · Cambridge, UK

- Somatic mutation signatures on DNA repair defective cell lines
 - Whole genome sequencing analysis of paired normal/tumor samples: SNV, Indel, CNVs
 - Whole exome sequencing analysis from TCGA (The Cancer Genome Atlas)
- Bioinformatics Analyst

2015

Congenica Ltd. · Cambridge, UK

- Splicing mutation effect analysis, mutation-created splice sites and cryptic splice sites
 - Bioinformatic analysis of trio and family from NGS data
- Bioinformatician — Molecular Pathology

2014 — 2015

Royal Marsden Hospital NHS Foundation Trust · London, UK

- NGS pipelines for tumor molecular pathology: panel targeted, immunoglobulin/TCR, amplicon analysis
 - Amplicon molecular barcoding deduplication pipeline
 - Study Bioinformatician at FoRMAT (molecular characterisation of advanced GI tumors)
- Bioinformatician — Clinical Neurogenetics

2013 — 2014

UCL / National Hospital for Neurology · London, UK

- NGS pipelines for clinical neurogenetics: exome, panel targeted, mtDNA heteroplasmy analysis
 - Database development for clinical information, diagnostic test results, and annotation
- Field Bioinformatics Specialist

2012 — 2013

Life Technologies · Mexico DF, Mexico

- Set up bioinformatic support and training for Ion Torrent (PGM/Proton) and SOLiD platforms across Latin America
 - Project design, troubleshooting for whole genome, resequencing, and targeted sequencing

Bioinformatician — Epigenetics & Cancer Biology

IDIBELL (Bellvitge Biomedical Research Institute) · Barcelona

- EPINORC project: Epigenetic Disruption of Non-Coding RNAs in Human Cancer (ERC Advanced Grant)
- Methylation arrays, bisulphite sequencing, WGS, exome NGS, RNA-seq, ChIP-seq analysis

Bioinformatics Research Assistant

2008 — 2010

CRG / Parc de Recerca Biomèdica de Barcelona

- De novo genome assembly, resequencing and genome analysis of *Saccharomyces cerevisiae*
- Developed DeathBase: relational database and web front-end for cell death curated proteins
- Characterization of putative human BH3-only proteins; cancer susceptibility gene analysis

Research Assistant

2005 — 2008

CSIC (CNB & CBM-SO) · Madrid

- Human centrosome proteome study and CentrosomeDB database development
- Cell death genetics in *Drosophila*; ML approach for phenotype trait analysis

 EDUCATION**Specialist — Quantitative Research & Statistics**

2006 — 2007

Polytechnic University of Madrid & CSIC

DEA / Master — Biochemistry & Molecular Biology

2005 — 2007

Universidad Autónoma de Madrid

B.Sc. Biochemistry

2001 — 2004

Universidad Autónoma de Madrid

 SELECTED PUBLICATIONS**PEER-REVIEWED PUBLICATIONS**

- Costanzo F, Martínez Díez M, Santamaría Nuñez G, Díaz-Hernández JI, Genes Robles CM, **Díez Pérez J**, Compe E, Ricci R, Li TK, Coin F, Martínez Leal JF, Garrido-Martín EM, Egly JM. "Promoters of ASCL1- and NEUROD1-dependent genes are specific targets of lurbectedin in SCLC cells." *EMBO Mol Med*, 2022. [PMID:35263037](#)
- Jiménez-Guardeño JM, Ortega-Prieto AM, Menéndez Moreno B, Maguire TJA, Richardson A, Díaz-Hernández JI, **Díez Perez J**, Zuckerman M, et al. "Drug repurposing based on a quantum-inspired method versus classical fingerprinting uncovers potential antivirals against SARS-CoV-2." *PLoS Comput Biol*, 2022. [PMID:35849631](#)
- Glodzik D, Morganella S, Davies H, ..., **Díez-Pérez J**, ..., Nik-Zainal S. "A somatic-mutational process recurrently duplicates germline susceptibility loci and tissue-specific super-enhancers in breast cancers." *Nature Genetics*, 2017. [PMID:28112740](#)
- Galindo-Moreno J, Iurlaro R, El Mjiyad N, **Díez-Pérez J**, Gabaldón T, Muñoz-Pinedo C. "Apolipoprotein L2 contains a BH3-like domain but it does not behave as a BH3-only protein." *Cell Death Dis*, 2014. [PMID:24901046](#)
- Heyn H, Li N, Ferreira HJ, Moran S, Pisano DG, Gomez A, **Díez J**, ..., Esteller M. "Distinct DNA methylomes of newborns and centenarians." *PNAS*, 2012. [PMID:22689993](#)
- Neurohr G, Naegeli A, Titos I, Theler D, Greber B, **Díez J**, Gabaldón T, Mendoza M, Barral Y. "A midzone-based ruler adjusts chromosome compaction to anaphase spindle length." *Science*, 2011. [PMID:21393511](#)
- **Díez J**, Walter D, Muñoz-Pinedo C, Gabaldón T. "DeathBase: a database on structure, evolution and function of proteins involved in apoptosis and other forms of cell death." *Cell Death Differ*, 2010. [PMID:20383157](#)
- Nogales-Cadenas R, Abascal F, **Díez-Pérez J**, Carazo JM, Pascual-Montano A. "CentrosomeDB: a human centrosomal proteins database." *Nucleic Acids Res*, 2009. [PMID:18971254](#)
- Bonifaci N, Berenguer A, **Díez J**, Reina O, Medina I, Dopazo J, Moreno V, Pujana MA. "Biological processes, properties and molecular wiring diagrams of candidate low-penetrance breast cancer susceptibility genes." *BMC Med Genomics*, 2008. [PMID:19094230](#)

PRESENTATIONS & CONFERENCES

- Parmar C, Roudko V, Muckerson V, Blando J, Daniel J, **Díez J**, Gunda S, Mendoza Alcala A, Franco A, Stambolic I, Benchea C, Stewart R, Lai Z, Reis-Filho J, Scaltriti M, Palmer D. "AI-Derived H&E Features Stratify Overall Survival and Suggest Differential Benefits in NEPTUNE (1L mNSCLC)." *AACR Annual Meeting*, San Diego, 2026
- Moorcraft SY, González de Castro D, Cunningham D, Walker BA, Hulkki Wilson S, **Díez Pérez J**, et al. "FoRMAT: Feasibility of a molecular characterization approach to treatment of patients with advanced GI tumors." *J Clin Oncol* 33, 2015 (ASCO). [DOI](#)
- McCall S, Polke JM, **Díez-Pérez J**, Gardiner A, Jaffer F, Nanji T, Pittman A, Hughes D, Hanna MG, Houlden H. "Next Generation Sequencing of Neuronal Channelopathies." *British Society for Genetic Medicine Annual Meeting*, Liverpool, 2013
- "ION PROTON DATA ANALYSIS" — Mexico & Latin America Ion Proton launch events, Life Technologies, 2012–2013
- ASHG Annual Meeting 2012, San Francisco · RECOMB::DREAM 2011, Bellvitge · Mono-allelic Expression in Health and Disease, Barcelona · Systems Biology & New Sequencing Technologies (SBNST), CRG Barcelona